

TARUVAR

Complete Architecture, Database Schema, Unique ID Generator,
Leadership Structure, Credentials & Operations Manual

A comprehensive reference guide for the Taruvar environmental movement (taruvar.org), covering the tree care verification engine, Supabase database, Unique ID card generation, dedicated page architecture, bottom navigation, and administrator protocols.

ORGANIZATION & DOMAIN

Taruvar Foundation (taruvar.org)

WHATSAPP HELPDESK

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GITHUB REPOSITORY

github.com/77naveenop/taruvar

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DOCUMENT VERSION & DATE

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1. Executive Summary & Core Mission

TARUVAR ("One Person. One Tree. / एक व्यक्ति, एक पेड़") is a technology-enabled grassroots environmental organization and public movement built on the core belief that **planting a tree is only the beginning**.

Traditional environmental drives focus solely on the gross number of saplings planted during one-day events, resulting in over 80% sapling mortality due to lack of ongoing care. Taruvar changes this through **individual guardianship (Paalna / देखभाल)**, requiring ongoing nurturing, photographic documentation, and a mandatory 5-month growth verification cycle.

The 5-Step Taruvar Framework

- 1. Plant (पौधा लगाना):** Individual plants a climate-appropriate sapling, selects species, and uploads photo proof.
- 2. Care (देखभाल / Paalna):** Guardian provides water, nutrients, and protection throughout seasonal heat and winter.
- 3. Document (तस्वीर व लॉग):** Adopter uploads monthly growth progress photos and observations.
- 4. Grow (निरंतर विकास):** Tree progresses through Month 1 to Month 5 verification milestones.
- 5. Inspire (प्रेरणा):** Verified trees earn the official *Level 5 Eco-Guardian Badge* and rank on the community feed.

2. System Architecture & Tech Stack

Layer	Technologies / Services	Description & Purpose
Frontend Framework	React 18, Vite 5, JavaScript (ESM)	High-performance SPA with instant navigation and zero routing lag.
Navigation UI	Sticky Bottom Navigation + Clean Top Header	Ergonomic 5-tab bottom navigation with center elevated Adopt pill button.
Dedicated Page Architecture	Standalone Full-Screen Pages	Dedicated routes for <code>/adopt</code> and <code>/auth</code> replacing popup modals.
Digital Identity Engine	Printable PVC Eco-Guardian ID Card	CR80 standard physical dimensions (85.6mm x 53.98mm) with QR code & print CSS.
Hero Animation	HTML5 Canvas 2D Engine (60 FPS)	Dynamic organic tree growth simulation from seed germination to full canopy.
Mobile PWA	Service Worker (<code>sw.js</code>), Web Manifest	Native 1-tap installation with safe-zone padded maskable icons.

Layer	Technologies / Services	Description & Purpose
Backend & Database	Supabase (PostgreSQL 15)	Relational data, user authentication, and row-level security (RLS).
Hosting & CI/CD	Vercel Edge Network	Automated continuous deployments triggered on git push to <code>main</code> branch.
DNS & Domain	GoDaddy DNS Registrar	Configured with Apex A record to Vercel IP (<code>216.198.79.1</code>) and CNAME for <code>www</code> .

3. Unique ID Generator & Printable PVC Card System

Taruvar issues official digital and physical credentials for every participant and adopted tree:

3.1 ID Hierarchy & Structure

Identifier	Format	Example	Purpose
Member ID	TRV-IND-YYYY-XXXX	TRV-IND-2026-4821	Permanent Eco-Guardian identity assigned to each registered caretaker.
Tree Passport ID	TRV-TREE-XXXX	TRV-TREE-8092	Unique serial code assigned to each adopted tree and its location.

3.2 Printable PVC Eco-Guardian Card Specifications

- **Standard Card Dimensions:** 85.6mm x 53.98mm (CR80 standard credit card / lanyard badge size).
- **Front Side:** Official Holographic Badge, Guardian Name, Member ID, Tree Passport ID, Species, Location, Live Scannable QR Code, Verification Status.
- **Back Side (Flip View):** 5-Month Milestone Verification Stamp Log (`Month 1` through `Month 5`) + Official Guardian Caretaker Oath.
- **Live Scannable QR Code:** Direct URL pointing to the tree's live public timeline (<https://taruvar.org/journey/TRV-TREE-XXXX>).
- **1-Click Physical Print:** Integrated with browser `@media print` CSS isolating the card for instant physical PVC card printing.

4. Database Architecture & Schema (Supabase)

4.1 Table: `pledges` (Tree Adoptions)

```
CREATE TABLE public.pledges (  
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,  
  user_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,  
  name TEXT NOT NULL,  
  email TEXT NOT NULL,  
  phone TEXT,  
  tree_type TEXT NOT NULL,          -- 'Neem', 'Peepal', 'Banyan', 'Mango', 'Guilmohar'  
  tree_name TEXT,                  -- Nickname e.g. 'My Peepal Guardian'  
  location TEXT,                   -- e.g. 'Sector 4 Green Park, Delhi NCR'  
  plantation_photo TEXT NOT NULL,  -- Photo proof  
  status TEXT DEFAULT 'pending',   -- 'pending', 'approved', 'rejected'  
  verified_months INT DEFAULT 1,   -- 1 to 5 milestones completed  
  upvotes_count INT DEFAULT 0,     -- Community ranking counter  
  member_id_code TEXT,             -- 'TRV-IND-2026-XXXX'  
  tree_id_code TEXT,               -- 'TRV-TREE-XXXX'  
  created_at TIMESTAMPTZ DEFAULT now()  
);  
  
ALTER TABLE public.pledges ENABLE ROW LEVEL SECURITY;  
CREATE POLICY "Public can view approved pledges" ON public.pledges FOR SELECT USING (true);  
CREATE POLICY "Authenticated users can insert pledges" ON public.pledges FOR INSERT WITH CHECK (tr  
CREATE POLICY "Admins can update pledges" ON public.pledges FOR UPDATE USING (true);
```

4.2 Table: `monthly_reports` (5-Month Growth Logs)

```
CREATE TABLE public.monthly_reports (  
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,  
  tree_id UUID REFERENCES public.pledges(id) ON DELETE CASCADE,  
  user_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,  
  month_number INT NOT NULL,       -- 1, 2, 3, 4, 5  
  photo_url TEXT NOT NULL,         -- Monthly growth photo  
  notes TEXT,                      -- Observations (height, compost, watering)  
  status TEXT DEFAULT 'pending',   -- 'pending', 'verified', 'rejected'  
  submitted_at TIMESTAMPTZ DEFAULT now()  
);  
  
ALTER TABLE public.monthly_reports ENABLE ROW LEVEL SECURITY;  
CREATE POLICY "Public read monthly_reports" ON public.monthly_reports FOR SELECT USING (true);  
CREATE POLICY "Users can insert monthly_reports" ON public.monthly_reports FOR INSERT WITH CHECK (
```

5. Core Leadership & Organizational Structure

Taruvar is guided by a 7-pillar foundational team:

Pillar	Leader Name	Designation & Role	Primary Responsibilities
Founder	Naveen Sharma	Founder & Lead Steward	Movement strategy, tech architecture, and nationwide environmental stewardship.
Botanical	Dr. Anita Verma	Botanical & Plantation Advisor	Indigenous sapling selection, organic soil nutrients, and survival protocols.
Operations	Rahul Mishra	Community Operations Lead	Neighborhood adoption circles, volunteer watering groups, and ward drives.
Green Shakti	Pooja Sundaram	Taruvar Green Shakti Lead	Women leadership initiatives and self-help group environmental collectives.
Youth & Campus	Aman Deep Singh	Youth & Campus Coordinator	University environmental chapters, student internships, and campus challenges.
Technology	Sneha Patel	Tree Journey Platform Lead	Digital tree tracker, QR passport engine, and community photo feed systems.
Partnerships	Vikram Joshi	Partnerships Coordinator	Corporate CSR alliances, school outreach, and institutional afforestation.

6. Authentication & Role-Based Access Control (RBAC)

Role	Who Receives It?	Capabilities & Permissions	Visibility
USER	Public users registering on <code>/auth</code>	Adopt trees, upload Month 1..5 logs, upvote feed, print ID card.	Standard UI (Admin Desk is 100% hidden)
ADMIN	Founder & Authorized Core Team Only	Approve tree adoptions, verify monthly reports, award badges.	Admin Verification Desk (Passkey Protected)

Founder Passkey: `TARUVAR_ADMIN_2026`

Entering this passkey in the Admin Verification Desk (accessible via footer link or `/admin`) permanently upgrades the user's role to `admin` in Supabase.

7. Official Contact Matrix & Credentials

Channel	Official Contact Detail	Notes / Purpose
Official Email	teamtaruvar@gmail.com	Primary inquiry and partner communication channel.
Voice Call Phone	+91 8543964107	Direct telephone support and coordinator line.
WhatsApp Helpdesk	+91 7887254107	Direct WhatsApp chat (https://wa.me/917887254107).
Instagram Official	@taruvarfoundation_	https://www.instagram.com/taruvarfoundation?stkn=dnIxa2prN2N0YXRi
Website Domain	https://taruvar.org	Hosted on Vercel Edge with GoDaddy DNS.

7.1 Environment Variables (Vercel & .env)

Variable Name	Config Location	Value / Description
VITE_SUPABASE_URL	Vercel Environment Variables	https://unvirsyjtuaajgnizuwto.supabase.co
VITE_SUPABASE_ANON_KEY	Vercel Environment Variables	[Configured in Vercel & .env (Supabase Public Anon Key)]
TARUVAR_ADMIN_PASSKEY	Application Secret	TARUVAR_ADMIN_2026

8. Step-by-Step Operational Workflows

8.1 Tree Adoption to Verified Badge Lifecycle

- Step 1 (Adoption):** User navigates to /adopt , chooses species, enters nickname/location, and uploads a mandatory plantation photo.
- Step 2 (ID Generation):** System immediately generates **Member ID** (TRV-IND-2026-XXXX) and **Tree Passport ID** (TRV-TREE-XXXX), rendering the printable PVC ID card.
- Step 3 (Admin Review):** Admin opens /admin , reviews the photo proof, and approves the adoption.
- Step 4 (5-Month Progress Logs):** Every 30 days, the user uploads Month 1 to 5 growth photos in their Profile.

5. **Step 5 (Verification & Badges):** Admin verifies monthly progress logs, advancing the user's status to *Level 5 Verified Eco-Guardian*.

8.2 Deploying Updates to Production (Vercel)

```
# Inside C:\Users\navvee\.gemini\antigravity\scratch\taruvar-website
git add .
git commit -m "Your description"
git push origin main
```

Vercel automatically builds and deploys the new release to **<https://taruvar.org>** within 45 seconds.